INSTALLATION & MAINTENANCE MANUAL

FOR

ELECTRO HYDRAULIC DOCK LEVELER

CLIENT : - L&T HYDRO CARBON

PROJECT : - IOCL

MFG.MONTH & YEAR : - OCT 2020

DESIGNED, DEVELOPED & MARKETED,

BY,

NIHVA TECHNOLOGIES PVT. LTD
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A. DESPATCH & UNLOADING: -

1. Loading for dispatch will be our responsibility. On receipt of the Dock-leveler at site, please check and ensure that there is no physical damage to the equipment.

2. While unloading the equipment from the transporting vehicle, please ensure to use slings with rated capacity higher than the gross weight of the Stationary lift in packed condition.

3. Slings MUST be hooked ONLY at specified positions. These positions are indicated and marked with a figure of 'HOOK'. Improper sling position or improper slinging could cause damage to the equipment & serious accidents.

B. ORDERING OF SPARES: -

1. When ordering for spare, please specify the following in your Purchase Order.

   - Equipment Name
   - Model Number
   - Equipment Serial No.
   - Part name / Description
   - Part Number
   - Quantity required
   - Delivery period / time
   - Mode of dispatch

   This would avoid ambiguity and help us serve you better.

DATASHEET FOR DOCK LEVELER: -

CLIENT: - L&T HYDROCARBON

1.0 CONSTRUCTION DETAILS

1.1 Required Capacity - each : 6 ton
1.2 Model No. : NDL5500
1.3 Leveler operation mode : Hydraulically with AC operated.
1.4 Lip operation mode : hydraulic
1.5 Operation control : by push buttons and control panel
1.6 Loading platform height from road Level : 1100 mm

1.7 Travel above & below dock level : (+) 6 degree / (-) 6-degree mm
1.8 Platform Size (L x W) : 5500 x 2000 mm
1.9 Lip size - (L x W) : 2000 x 630 mm
1.10 Working length : 5500 mm
1.11 Overall length : 6130 mm
1.12 Maximum slope can be achieved : 8 to 10%
1.13 Clear Pit dimensions (L x W X D) : 5500 x 2050 x 805 mm
1.14 Hydraulic power pack
- 1.14.1 Driver type : Direct by electric Motor
- 1.14.2 Absorbed power : 0.75 kW
- 1.14.3 FLP Motor Ratings : 0.75/1 kW /HP
- 1.14.4 Degree of protection for motor: IP 55
- 1.14.5 Motor RPM : 1410
- 1.14.6 Pump design : 1.6 cc/rev.
- 1.14.7 Pump pressure : 200 (max) Kg/cm²g
- 1.14.8 Safety valve set pressure : 90 Kg/cm²g
- 1.14.9 Pump type : Gear
- 1.14.10 Valves required : By vendor
- 1.14.11 Volume of oil receiver : 5 Lit
- 1.14.12 First fill of oil / Qty : By Vendor
- 1.14.13 Hydraulic tubing : By Vendor
- 1.14.14 Instrumentation : By vendor
- 1.14.15 Control panel : By vendor

1.15 Empty weight : APPROX. 3000 Kg

DOCK LEVELLER

INTRODUCTION: -

The hydraulic Dock Leveler is the result of many years' experience in materials handling engineering. "NIHVA" offers the best possible factor of safety with simple, easy to use controls. The high quality of manufacture gives a robust construction and many years of useful life. Each lip and platform movement can be individually controlled (or corrected) by push button.

The dock leveler can be set at any position. During loading, the Dock Leveler is in floating mode i.e. it follows the up/down movements of the vehicle deck. Even with lopsided vehicles, the Dock leveler lip will remain in contact, and toe-traps are avoided by the tilt built into the Dock Leveler.

The whole platform is designed to flex laterally adequately to adapt to sideways tilted vehicles. The leveler always seals perfectly to the lorry bed.

PRINCIPLE: -

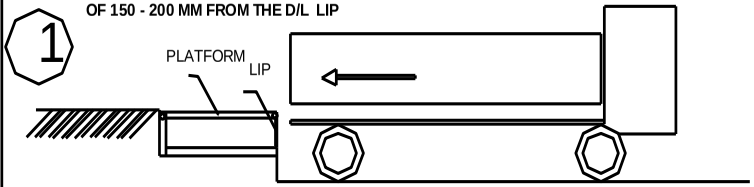
Dock Leveler is an adjustable platform used between the transport vehicle and dispatch area of the warehouse of factory for efficient and safe material transfer applications. It bridges the gap between the dispatch area and the transport vehicle and compensates for variations in the truck bed height.

PRE – OPERATING INSTRUCTIONS: -

1. Ensure Hydraulic Oil in the power pack.
2. Ensure Motor connection and proper earthing.
3. Ensure joint leakage.
4. Ensure proper location of platform and lip.

1

REVERSE THE TRUCK UPTO DISTANCE
OF 150 - 200 MM FROM THE D/L LIP

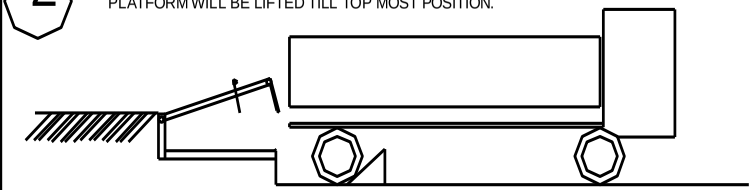


NOTE : ENSURE THAT EDGES OF TRUCK LOAD BODY ARE OUTSIDE LIP WIDTH
I.E. CHECK ALIGNMENT OF LIP WITH TRUCK BODY.

2

HOLD PLATFORM UP BUTTON IN PRESSED
POSITION.

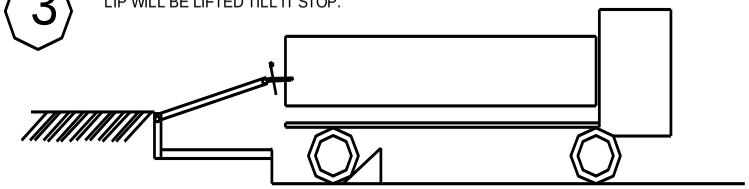
PLATFORM WILL BE LIFTED TILL TOP MOST POSITION.



3

HOLD LIP UP BUTTON IN PRESSED POSITION.

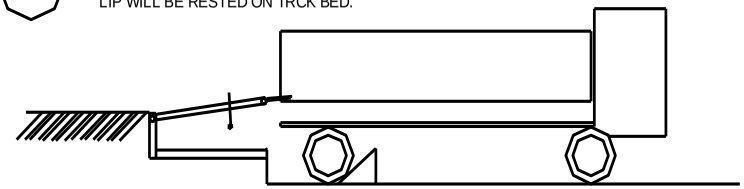
LIP WILL BE LIFTED TILL IT STOP.



4

HOLD PLATFORM DOWN BUTTON IN PRESSED
POSITION.

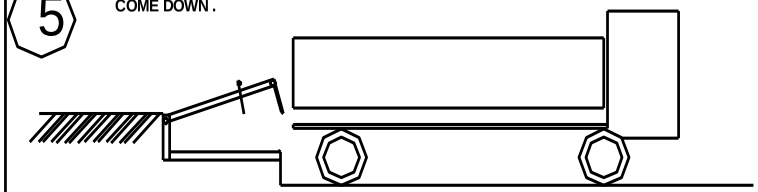
LIP WILL BE RESTED ON TRCK BED.



UNLOAD OR LOAD THE MATERIAL FROM THE TRUCK.

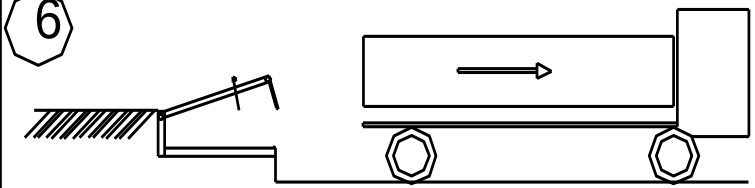
5

PRESS PLATFORM UP TO CLEAR THE LIP
FROM TRUCK BED THAT TIME THE LIP WILL
COME DOWN .



6

DRIVE THE TRUCK AWAY FROM D/L.



7

HOLD PLATFORM DOWN BUTTON .

LIP WILL BE PARKED TO ITS ORIGINAL POSITION.



OPERATING INSTRUCTION FOR DOCK- LEVELER.

NIHVA TECHNOLOGIES PVT. LTD.

ROUTINE MAINTENANCE AND LUBRICATION

These recommendations are applicable to the Dock Leveler considering average daily usage of 15 to 20 times. If it is used more frequently, the same time intervals for the maintenance should be adjusted accordingly.

DAILY: -

1. Check Hydraulic pipe lines for leak proof working.
2. Check electrical connections for their proper working.

MONTHLY: -

1. Grease the hinge pins at bottom frame and top plate joint and extension lip plate hinge joint.
2. Grease the main Jack Cylinder and Lip Jack Cylinder, Top and Bottom pins if some frictional sound hears.
3. Change Hydraulic Oil (servo 68) viscosity from hydraulic expert & change if required.

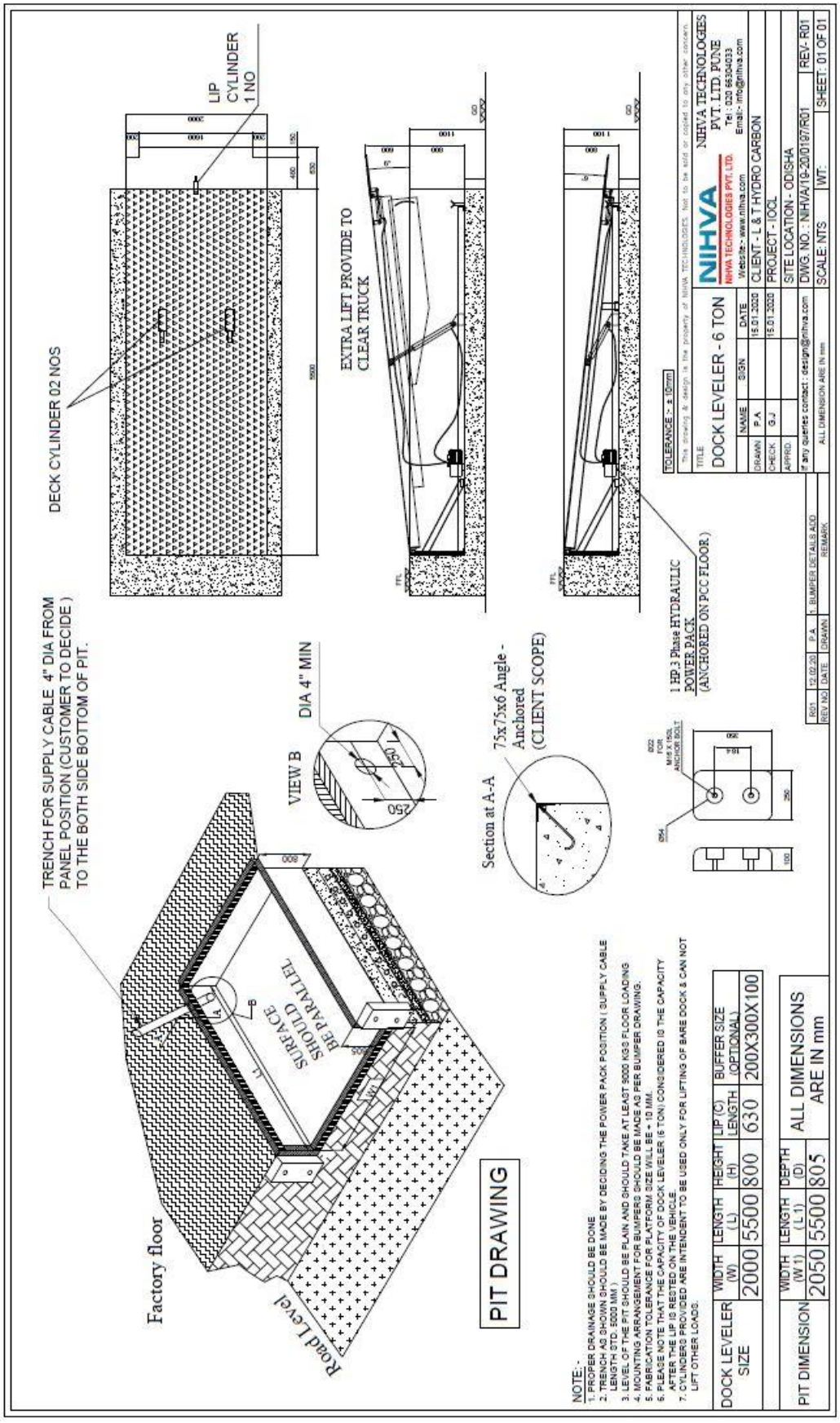
ASSEMBLIES: -

1. GA DRAWING.
2. DECK CYLINDER DETAILS.
3. LIP CYLINDER DETAILS.
4. HYDRAULIC POWER PACK DETAILS & CIRCUIT.
5. ELECTRICAL CIRCUIT.

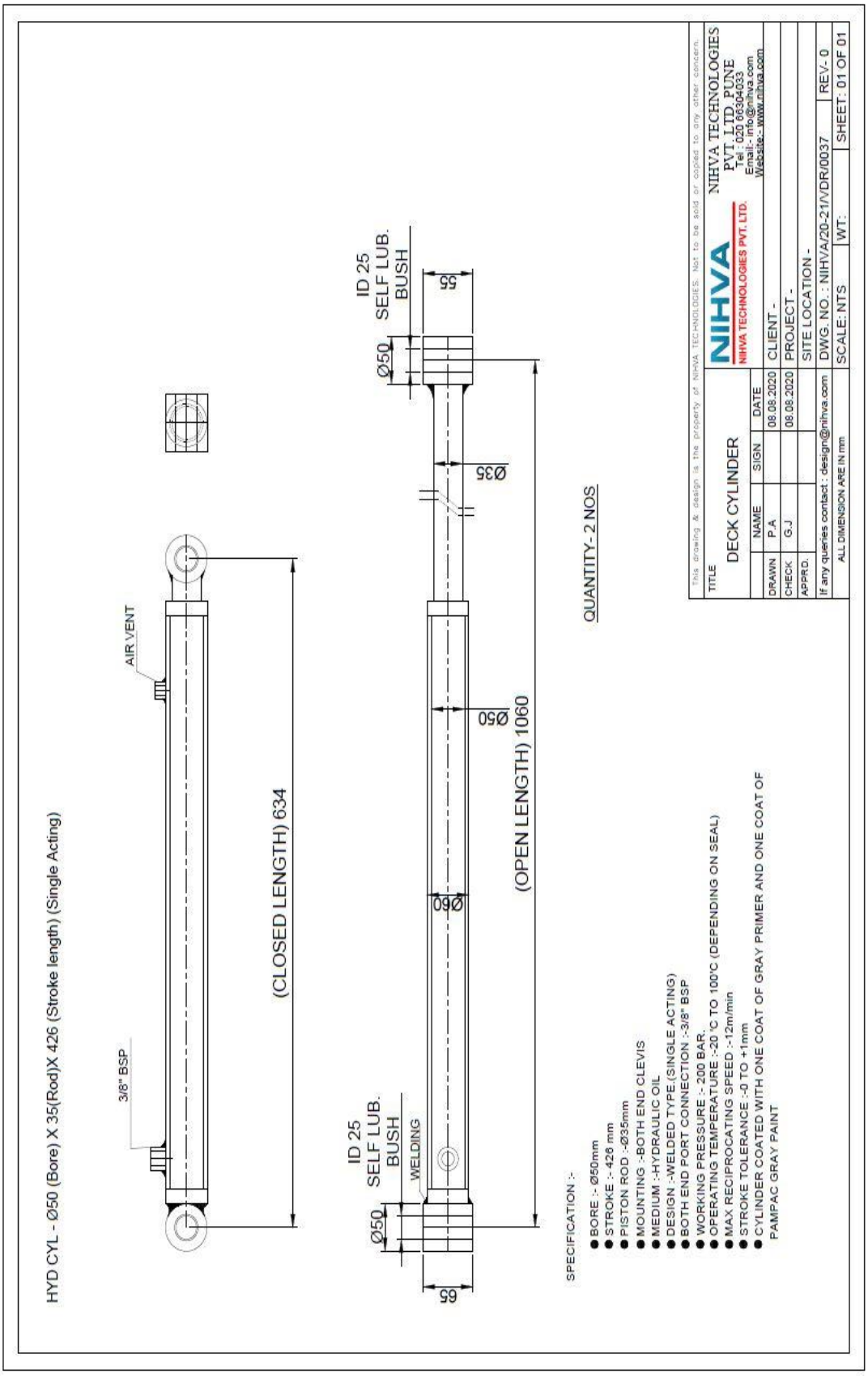
SPARE PARTS LIST: -

1. SIEMENS FLANGE MOUNTED MOTOR

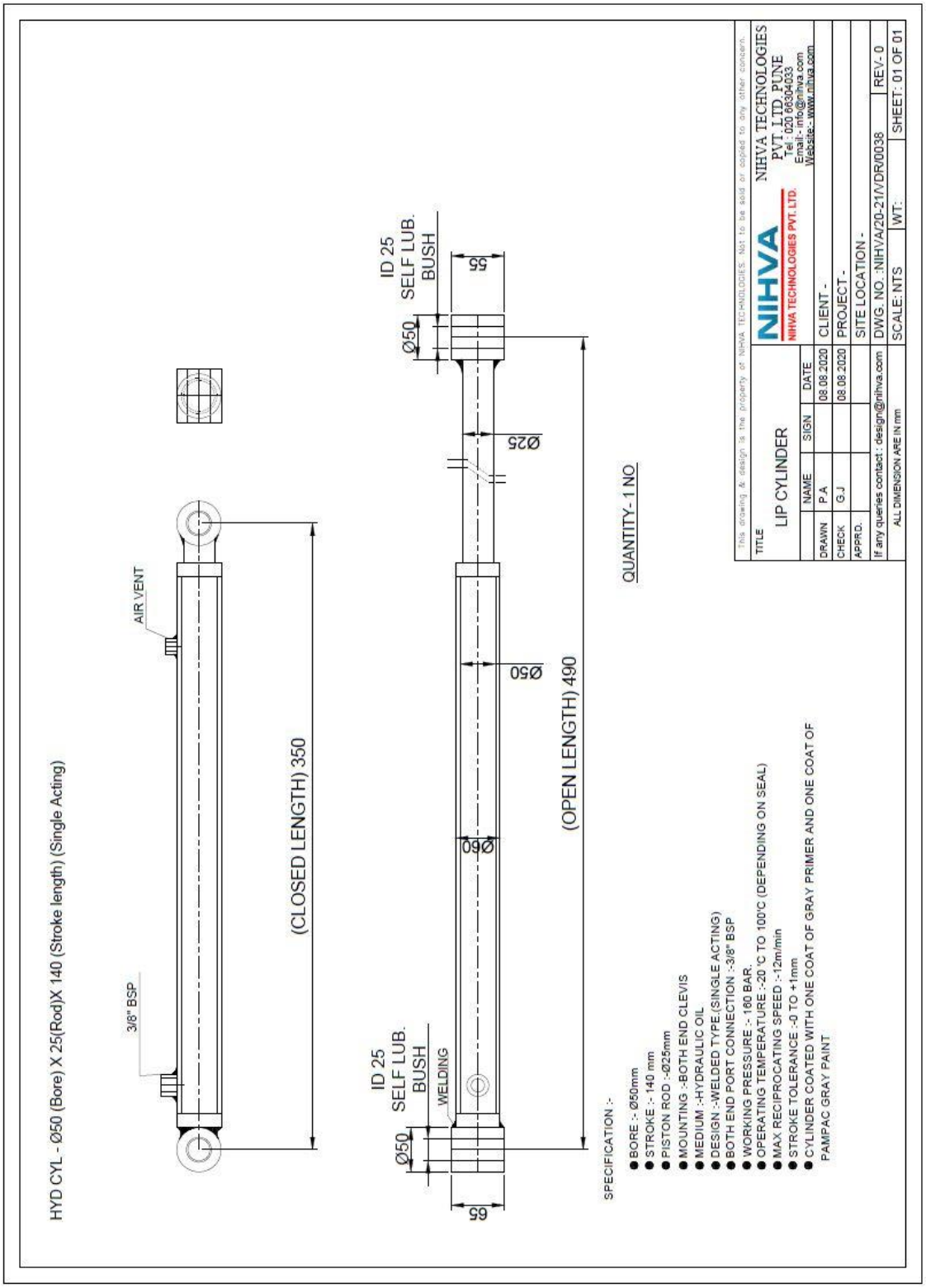
GA DRAWING:-


LIP CYLINDER 1 NO
DECK CYLINDER 02 NOS
EXTRA LIFT PROVIDE TO CLEAR TRUCK
TRENCH FOR SUPPLY CABLE 4" DIA FROM PANEL POSITION (CUSTOMER TO DECIDE) TO THE BOTH SIDE BOTTOM OF PIT.
VIEW B
DIA 4" MIN
Section at A-A
75x75x6 Angle - Anchored (CLIENT SCOPE)
1 HP,3 Phase HYDRAULIC POWER PACK (ANCHORED ON PCC FLOOR)
Factory floor
Road Level
SURFACE SHOULD BE PARALLEL
PIT DRAWING
NOTE:-
1. PROPER DRAINAGE SHOULD BE DONE
2. TRENCH AS SHOWN SHOULD BE MADE BY DECIDING THE POWER PACK POSITION (SUPPLY CABLE LENGTH STD. 5000 MM)
3. LEVEL OF THE PIT SHOULD BE PLAIN AND SHOULD TAKE AT LEAST 5000 KG/S FLOOR LOADING
4. MOUNTING ARRANGEMENT FOR BUMPERS SHOULD BE MADE AS PER BUMPER DRAWING.
5. FABRICATION TOLERANCE FOR PLATFORM SIZE WILL BE + 10 MM.
6. PLEASE NOTE THAT THE CAPACITY OF DOCK LEVELER (6 TON) CONSIDERED IS THE CAPACITY AFTER THE LIP IS RESTED ON THE VEHICLE.
7. CYLINDERS PROVIDED ARE INTENDENT TO BE USED ONLY FOR LIFTING OF BARE DOCK & CAN NOT LIFT OTHER LOADS.
DOCK LEVELER SIZE: WIDTH (W) 2000 | LENGTH (L) 5500 | HEIGHT (H) 800 | LIP (C) LENGTH 630 | BUFFER SIZE (OPTIONAL) 200X300X100
PIT DIMENSION: WIDTH (W1) 2050 | LENGTH (L1) 5500 | DEPTH (D) 805 | ALL DIMENSIONS ARE IN mm
REV NO 001 | DATE 12.02.20 | DRAWN P.A. | REMARK: R. BUMPER DETAILS ADD
TOLERANCE :- ± 10mm
This drawing & design is the property of NIHVA TECHNOLOGIES. Not to be sold or copied to any other person.
TITLE: DOCK LEVELER - 6 TON
NAME | SIGN | DATE
DRAWN P.A. | 15.01.2020
CHECK G.J | 15.01.2020
APPRD.
If any queries contact : design@nihva.com
ALL DIMENSION ARE IN mm
NIHVA TECHNOLOGIES PVT. LTD. PUNE
Tel : 020 66304033
Email:- info@nihva.com
Website :- www.nihva.com
CLIENT - L & T HYDRO CARBON
PROJECT - IOCL
SITE LOCATION - ODISHA
DWG. NO. : NIHVA/19-20/0197/R01 | REV- R01
SCALE: NTS | WT: | SHEET: 01 OF 01

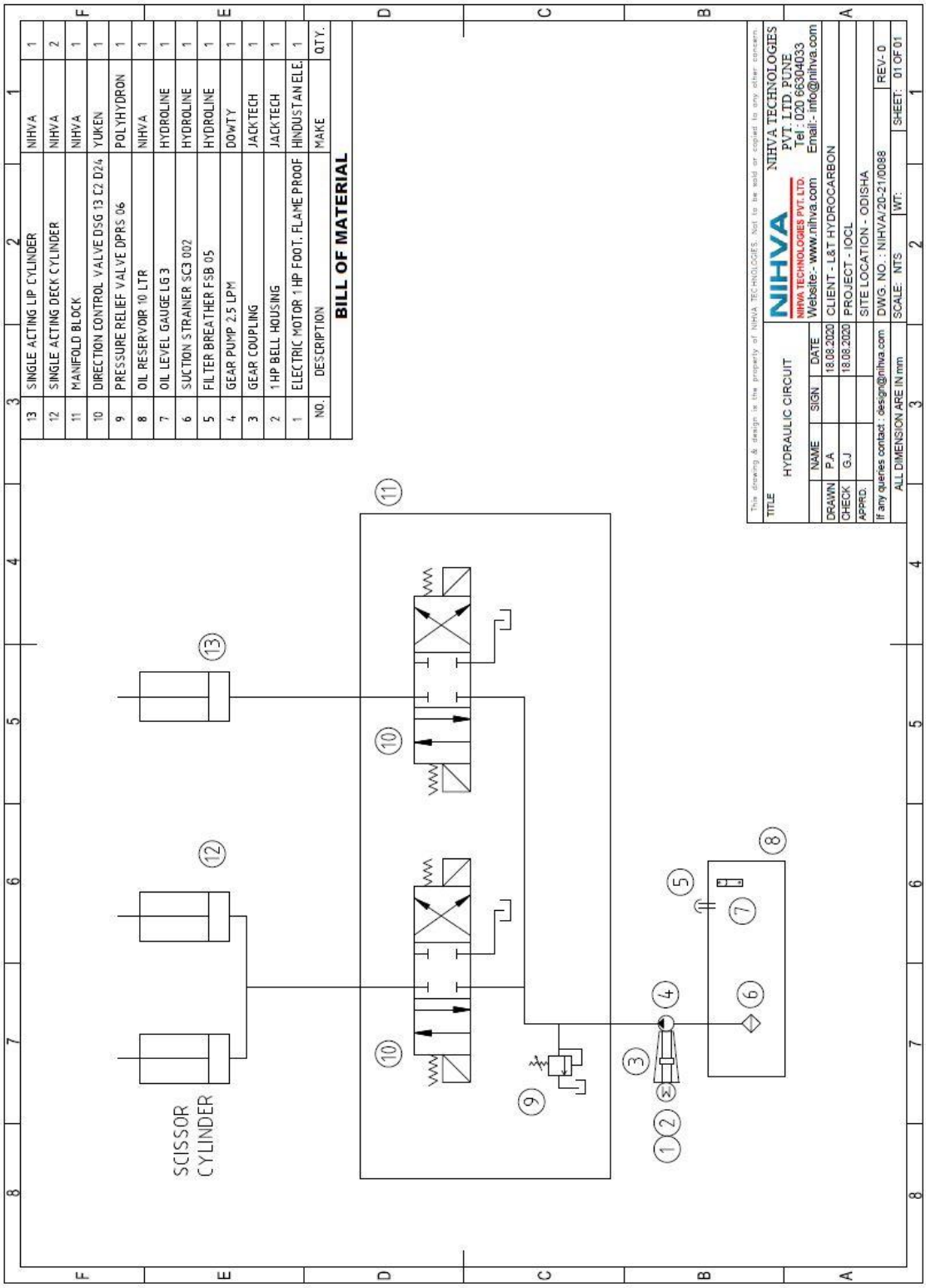
DECK CYLINDER:-



LIP CYLINDER :-



HYDRAULIC CIRCUIT:–



ELECTRIC CIRCUIT & PART LIST:-

